CSE 219 — Signals and Linear Systems

Lec 5–6 · Fourier Series and Its Properties

Lesson notes · Exam approach · TF practice with solutions · Quick notes (Anik Sir's portion)

How to Read These Notes

EXAM FOCUS	Exam importance, where it appears in the TF collection
MEMORIZE	Rules, tables and formulas to memorize
TRAP	Common mistakes and exam traps
EXAM-READY ANSWER	Model wording you can write in the exam
NOTE	Extra explanation / clarification
Grey box, monospace	Code, worked tables, derivations and hand-drawn sketches
Centered grey box	Formula to remember / final numerical result
Purple heading / ✓ green	Practice questions / their solutions

Contents

Exam Map — What the TF Papers Ask — Fourier-series questions and their weight
Lesson 11 — The Idea of Fourier Series; Orthogonality and Convergence — harmonics, the coefficient formula, orthogonal sets, Dirichlet
Lesson 12 — Computing Coefficients — by inspection, by integration; square wave, ramp; trigonometric form
Lesson 13 — Linearity, Shift, Reversal, Scaling and Symmetry — coefficient rules; real/even/odd/half-wave
Lesson 14 — Differentiation, Impulses, Impulse Trains, Integration — $jk\omega_0$ rule, impulse train $a_k = 1/T$, sawtooth
Lesson 15 — Multiplication, Parseval, and Periodic Inputs to LTI Systems — coefficient convolution; $y_k = H(jk\omega_0)a_k$
Quick Notes — Fourier Series — merged revision notes + key diagrams

NOTE Slide numbers refer to Lecture-5-Fourier-Series-Let's-Look-Inside.pdf (53 slides, “Lec 5”) and Lecture-6-Properties-of-Fourier-Series.pdf (82 slides, “Lec 6”). Orthogonal function sets, Dirichlet conditions, half-wave symmetry, Parseval's relation and the response of an LTI system to a periodic input are textbook background (marked); the EEE and CSE 311 papers ask them often.

Exam Map — What the TF Papers Ask

Fourier series appear in every paper: a full 30-mark question in CSE 219 2023-24 (Q4) and two to three parts in every EEE paper.

Session (paper)	Questions answered by this note
2023-24 (CSE 219)	Q4(a) Odd square wave (±A), y = x(t−T_0/6): half-wave symmetry? (5) · Q4(b)–(d) Y_k of the shifted square wave; magnitude/phase spectra to k=5; compare with trig FS (25)
2023-24 (EEE 211)	Q7(a) Three functions pairwise orthogonal on (−1,2); A for orthonormal (10) · Q7(b) Half-wave rectified sine (T=0.5 ms) → differentiator & LPF (1 kHz, gain 5): FS of y_1, y_2(t) (25)
2022-23 (CSE 219)	Q5(d) x(t) = $\Sigma_{k=0}^{100}$ (½)^k $e^{jk2\pi t/50}$: real-valued? even? (5)
2022-23 (EEE 211)	Q5(a) Compact trig FS of 2+3cos2t+4sin2t+2sin(3t+30°)−cos(7t+150°); amplitude spectrum (10) · Q7(a) Exponential FS of the periodic waveform of Fig. 7(a) (20) · Q8(a) Trig FS of the impulse train δ_{T0}(t); amplitude spectrum (20)
2022-23 (CSE 311)	Q1(a) FS coefficients: even signal → real, odd → imaginary; cos $2\pi f_0 t$ (17)
2021-22 (EEE 211)	Q3(a) Periodic x (T=4) = δ(t−1)+2[u(t−2)−u(t−4)] into LC–R circuit: spectra of x and y (20) · Q3(b) Orthogonal set on (−2,2); orthonormalise; expand x = 1 on [0,2]; Parseval (15)
2020-21 (EEE 211)	Q7(a) Triangle wave x: exp FS; y (rect train) via FS properties; amplitude spectra (20) · Q7(b) y''+3y'+2y = x'+x, x = 2cos t + 6cos 3t: spectra of y; average power (15)
2020-21 (CSE 311)	Q2(b) Exponential FS coefficients of sin 2πt and cos 2πt; need a full cycle? (17)
2019-20 (EEE 211)	Q5(b) Full-wave rectified sin t into RC low-pass (15 Ω, 1/5 F): H(ω), y(t) (20) · Q7(a) Exponential FS of the two impulse trains of Fig. 7(a) (15) · Q8(b) Trig FS a_0, a_n, b_n of the triangular wave of Fig. 8(b) (20)
2019-20 (CSE 311)	Q1(a) Why cos/sin $2\pi f_0 t$ has a FS; are its exp. FS coefficients real or imaginary? (17)
2018-19 (EEE 211)	Q1(a) Orthogonality condition; e^{j2t}, e^{j4t} orthogonal on [0, π]? (10) · Q1(b) x = 2+½cos(t+45°)+2cos3t−sin(4t+30°): exp FS; magnitude/phase spectra (10) · Q1(c) Triangular wave (T=4, A=4) into R = 100 Ω + L = 0.1 H: power in R (15) · Q3(c) C_n = 1 (odd n), T = 2, through band-pass H with ±π/2 phase: h(t), output (15)
2018-19 (CSE 311)	Q2(b) Even → real, odd → imaginary FS coefficients; half-cycle integration (15)
2017-18 (EEE 211)	Q1(a) Orthogonal signals; show φ_1, φ_2, φ_3 orthogonal; T for orthonormal (12) · Q1(c) H = 1 for |ω| ≤ 3π; x = 1+2cos2πt+cos4πt+½cos6πt: exp FS, y, spectra (15) · Q4(a) Trig FS coefficients of 3 + cos(4t+π/4) + sin(10t+π/3) (12)
2016-17 (EEE 211)	Q2(a) Dirichlet conditions for FS convergence (10) · Q2(b) Pulse train: exp FS, power (Parseval), FS of dx/dt, output of 1/(1+jω), spectra (25) · Q3(a) Orthogonal set; T for orthonormal; express x = A on (0,T) (15)
2015-16 (EEE 301)	Q4(a) Periodic x (FS given) into R–L–C circuit: y(t); distortionless? (20) · Q4(b) x = cos4πt, y = sin4πt (T = ½): FS of z = xy by multiplication property (15) · Q6(a) Impulse train: FS, FT and power spectral density (10)
2014-15 (EEE 301)	Q4(a) C_n = ½[1−(−1)^n], T = 20 ms: real signal? time-domain signal (18) · Q4(b) C_n = $e^{jn\pi/3}$/(n^2+1) into L–C–R circuit: first four harmonics, average of output (17)
2013-14 (EEE 301)	Q3(a) Do tan(2πt) and sin(2π/t) have convergent FS? (18) · Q3(b) Exponential FS of the triangular wave (peak 2, period π) (17) · Q4(b) x = B + A cos(ω_0t+θ) into L–C–R circuit: y(t) (17)
2012-13 (EEE 301)	Q4(a) Express x(t) of Fig. 4(a) in the basis φ_1, φ_2, φ_3 on [0,3] (17) · Q4(b) Half-wave rectified cos t into L–C–R circuit: 2nd and 4th harmonics of y (18)
2011-12 (EEE 301)	Q1(a) Real odd, T = 2, c_k = 0 for |k|>1, ½∫|x|² = 1: two signals (18) · Q1(b) H = sin4ω/ω, periodic square input: output by FS (17)
2010-11 (EEE 301)	Q1(a) Square wave K, width τ, period T: C_n; spectra for τ=1, T=5,10,15 (18) · Q1(b) x = A e^{j10^3t} into RC circuit: H, magnitude/phase, y(t) (17)

EXAM FOCUS Most asked: (1) coefficients by inspection from a sum of sinusoids (EEE every year, 10–17 marks) · (2) coefficients of a standard periodic waveform (square, triangle, rectified sine, impulse train) · (3) use properties instead of integrating — shift, derivative (CSE 219 2023-24 Q4, EEE 2020-21 Q7(a)) · (4) periodic input through an LTI filter: y_k = H(jkω_0)a_k (EEE 2023-24 Q7(b), 2019-20 Q5(b)) · (5) even → real, odd → imaginary coefficients (CSE 311 three times).

Lesson 11 — The Idea of Fourier Series; Orthogonality and Convergence

Slides 2–23, 44–52 (Lec 5) · orthogonal sets and Dirichlet conditions: textbook

1. Periodic Signals as Sums of Harmonics (sl. 2–17)

Any periodic signal can be split into simple waves; each simple wave is easy to handle (a vibrating string, heat diffusion), and the answers are added back. A signal of period T contains only frequencies $k\omega_0$, $\omega_0 = 2\pi/T$ (bells ringing every 2, 3, 5 min repeat every lcm = 30 min, sl. 10–11).

$$x(t) = \Sigma_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \qquad \omega_0 = 2\pi/T$$

2. Finding a_k: Stop the Rotation and Average (sl. 18–22)

The average of $e^{jk\omega_0 t}$ over one period is 1 for k = 0 and 0 otherwise. Multiply x(t) by $e^{-jk\omega_0 t}$: the k-th term stops rotating, every other term still rotates and averages to zero.

$$a_k = (1/T) \int_T x(t)\, e^{-jk\omega_0 t}\, dt \qquad a_0 = \text{average value of } x$$

3. Coordinates and Projections (sl. 44–52)

Functions have coordinates just like vectors: 3 + 2t ↔ (3, 2) in the basis {1, t}; e^t ↔ (1, 1, 1/2!, …) in {1, t, t^2, …}. Fourier series uses the basis $\{e^{jk\omega_0 t}\}$, and the coordinate is a projection with the inner product $\langle x, z\rangle = (1/T)\int_T x\, z^*\, dt$. With $z = e^{jk\omega_0 t}$ this is exactly a_k.

4. Orthogonal and Orthonormal Sets (textbook)

- φ_i, φ_j are orthogonal on (t_1, t_2) if $\int \varphi_i \varphi_j^*\, dt = 0$ $(i \neq j)$; orthonormal if also $\int|\varphi_i|^2\, dt = 1$.
- Normalize: divide φ_i by $\sqrt{E_i}$, $E_i = \int|\varphi_i|^2$.
- Expansion: $x \approx \Sigma\, c_i\, \varphi_i$ with $c_i = \int x\, \varphi_i^*\, dt / E_i$ (best least-squares coefficients).
- Parseval for the set: $\int|x|^2\, dt = \Sigma\, |c_i|^2\, E_i$ when the set is complete for x.

5. Dirichlet Conditions (textbook)

A periodic x(t) has a Fourier series that converges to x(t) at every point of continuity (and to the midpoint of each jump) if, over one period: (1) it is absolutely integrable, $\int_T|x|dt < \infty$; (2) it has a finite number of maxima and minima; (3) it has a finite number of finite discontinuities. They are **sufficient, not necessary**: signals that violate them may still have useful series/transforms (e.g. impulse trains, u(t), sinusoids for the FT). Near a jump the partial sums overshoot by about 9% (Gibbs).

TRAP • tan(2πt) violates condition 1 (infinite jumps, not integrable); sin(2π/t) has infinitely many oscillations near t = 0 (and is not even periodic). • Orthogonality must be checked on the stated interval only.

Exam Approach

1. Orthogonality: compute every pairwise integral over the interval (disjoint supports give 0 at once).
2. Normalize with A = 1/√(energy).
3. Expansion: $c_i = \langle x, \varphi_i\rangle / E_i$, then check a value or Parseval.
4. Dirichlet: test each of the three conditions for one period and state which fails.

Practice — Lesson 11

(a) Actual — EEE 211 2023-24 Q7(a). (b) Actual — EEE 211 2018-19 Q1(a). (c) Actual — EEE 301 2012-13 Q4(a). (d) Actual — EEE 301 2013-14 Q3(a), CSE 311 2020-21 Q3(b).

(a) $\varphi_1 = -A$ on $(-1, 0)$, A on $(0, 1)$; $\varphi_2 = A$ on $(-1, 1)$; $\varphi_3 = 1$ on $(1, 2)$. Show they are pairwise orthogonal on $(-1, 2)$ and find A for orthonormality. (10)

(b) State the condition for orthogonality. Are e^{j2t} and e^{j4t} orthogonal on $[0, \pi]$? (10)

(c) $\varphi_1 = 1$ $(0 \le t \le 2)$, $\varphi_2 = 1$ (0..1), -1 (1..2), $\varphi_3 = 1$ (2..3). Express x(t) = 2 (0..1), −1 (1..2), 1 (2..3).

(d) Do tan(2πt) and sin(2π/t) have convergent Fourier series? State the Dirichlet conditions; are they necessary?

✓ Solution

(a) $\int\varphi_1\varphi_2 = A^2[\int_{-1}^{0}(-1)dt + \int_0\text{^}1\ 1\ dt] = 0$; φ_3 is zero wherever φ_1, φ_2 are nonzero $\Rightarrow \int\varphi_1\varphi_3 = \int\varphi_2\varphi_3 = 0$. Energies: $\int\varphi_1^2 = \int\varphi_2^2 = 2A^2$, $\int\varphi_3^2 = 1$. Orthonormal $\Rightarrow 2A^2 = 1 \Rightarrow$ **$A = 1/\sqrt{2}$** (φ_3 already has unit energy).

(b) $\int_0\text{^}\pi\ e^{j2t}(e^{j4t})^*\ dt = \int_0\text{^}\pi\ e^{-j2t}dt = (1 - e^{-j2\pi})/(2j) = 0 \Rightarrow$ **orthogonal**. (Each has energy π, so $\sqrt{(1/\pi)}e^{j2t}$, $\sqrt{(1/\pi)}e^{j4t}$ are orthonormal.)

(c) $E_1 = 2$, $E_2 = 2$, $E_3 = 1$. $c_1 = (2 - 1)/2 = ½$; $c_2 = (2\cdot1 + (-1)(-1))/2 = 3/2$; $c_3 = 1$. $x = ½\varphi_1 + (3/2)\varphi_2 + \varphi_3$ (check (1,2): ½ − 3/2 = −1 ✓).

(d) tan(2πt): period ½ but $|x| \to \infty$ at $t = ¼ + k/2$ — not absolutely integrable ⇒ no convergent FS. sin(2π/t): not periodic, and has infinitely many maxima/minima near t = 0 ⇒ fails condition 2. Dirichlet conditions as in §5; sufficient but not necessary (cos $\omega_0 t$, impulse trains have series/transforms in the generalized sense).

Lesson 12 — Computing Coefficients

Slides 24–43 (Lec 5) · trigonometric form: textbook

1. By Inspection (sl. 25–29)

If x is already a sum of sinusoids, use Euler: $\cos\theta = (e^{j\theta} + e^{-j\theta})/2$, $\sin\theta = (e^{j\theta} - e^{-j\theta})/2j$.

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x = 1 + sin ω₀t + 2cos ω₀t + cos(2ω₀t + π/4)
a₀ = 1;  a₁ = 1 + 1/(2j) = 1 − j/2;  a₋₁ = 1 + j/2;  a₂ = ½e^{jπ/4} = (√2/4)(1 + j);  a₋₂ = (√2/4)(1 − j)
|a₁| = √5/2, ∠a₁ = −tan⁻¹(½);  |a₂| = ½, ∠a₂ = π/4
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2. Square Wave (sl. 30–34)

x = 1 for $|t| < T_1$, 0 for $T_1 < |t| < T/2$:

$$\mathbf{a_0 = 2T_1/T} \qquad \mathbf{a_k = \sin(k\omega_0 T_1)/(k\pi)} \quad \mathbf{(k \neq 0)}$$

$T_1 = T/4$: $a_0 = ½$, a_k = sin(kπ/2)/(kπ): $a_{\pm1} = 1/\pi$, $a_{\pm3} = -1/(3\pi)$, even k = 0.

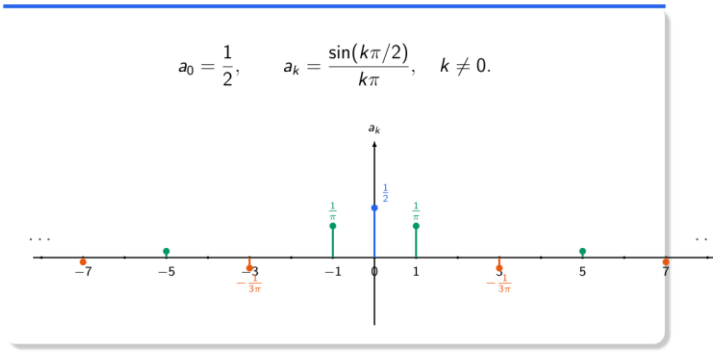


Figure 12.1 — Coefficients of the square wave with $T_1 = T/4$ (Lec 5, sl. 34)

3. Triangular Ramp, T = 3 (sl. 35–41)

x = t (0 ≤ t < 1), (3 − t)/2 (1 ≤ t < 3). a_0 = area/period = (½·3·1)/3 = ½ (geometry, sl. 37). For k ≠ 0 the integrals give $(e^{-jk\omega_0} - 1)/(2k^2\omega_0^2)$.

TRAP Slide 39 has a sign slip in the second integral, so slides 40–41 print a_k = $(e^{-jk\omega_0} - 1)/(6k^2\omega_0^2)$. The correct value is $(e^{-jk\omega_0} - 1)/(\mathbf{2}k^2\omega_0^2)$, $\omega_0 = 2\pi/3$ (checked numerically; quickest derivation: x″ has impulses +3/2 at t = 0 and −3/2 at t = 1 per period, so $(jk\omega_0)^2 a_k = (1/3)(3/2)(1 - e^{-jk\omega_0})$).

4. Trigonometric Forms (textbook)

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x(t) = a₀ + Σ_{n≥1} [aₙ cos nω₀t + bₙ sin nω₀t]        aₙ = (2/T)∫x cos nω₀t dt,  bₙ = (2/T)∫x sin nω₀t dt
     = C₀ + Σ Cₙ cos(nω₀t + θₙ)   (compact)            Cₙ = √(aₙ² + bₙ²),  θₙ = −tan⁻¹(bₙ/aₙ)
Exponential ↔ trig:  c₀ = a₀;  cₙ = (aₙ − jbₙ)/2;  c₋ₙ = cₙ*;  Cₙ = 2|cₙ|
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Exam Approach

1. Sum of sinusoids → inspection; rewrite sines as cosines with phase, collect same frequencies.
2. Standard waveform → integrate over the most convenient period; k = 0 separately.
3. Piecewise-linear waveform → differentiate twice to get impulses (Lesson 14), then divide by $(jk\omega_0)^2$.

4. Spectra: plot |a_k| (even) and ∠a_k (odd) against k or $k\omega_0$.

Practice — Lesson 12

(a) Actual — EEE 211 2022-23 Q5(a). (b) Actual — EEE 211 2018-19 Q1(b). (c) Actual — CSE 311 2019-20 Q1(a), 2020-21 Q2(b). (d) Actual — EEE 211 2019-20 Q8(b). (e) Older: EEE 211 2017-18 Q4(a), EEE 301 2013-14 Q3(b), 2010-11 Q1(a).

(a) Compact trigonometric FS and amplitude spectrum of x = 2 + 3cos2t + 4sin2t + 2sin(3t + 30°) − cos(7t + 150°). (10)

(b) Exponential FS and magnitude/phase spectra of x = 2 + ½cos(t + 45°) + 2cos3t − sin(4t + 30°). (10)

(c) Exponential FS coefficients of cos $2\pi f_0 t$ and sin $2\pi f_0 t$: real or imaginary? Must we integrate over a full cycle? (17)

(d) Triangular wave, period 2: x(0) = −A, x(±1) = A, linear in between (zero crossings at ±½). a_0, a_n, b_n. (20)

(e) (i) 3 + cos(4t + π/4) + sin(10t + π/3) (trig coefficients); (ii) triangle wave rising from 0 at t = 0 to 2 at t = π/2 and back to 0 at t = π, period π; (iii) square wave K of width τ, period T: C_n and the effect of T.

✓ Solution

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(a) ω₀ = 1.  3cos2t + 4sin2t = 5cos(2t − 53.13°);  2sin(3t + 30°) = 2cos(3t − 60°);
    −cos(7t + 150°) = cos(7t − 30°)
    x = 2 + 5cos(2t − 53.13°) + 2cos(3t − 60°) + cos(7t − 30°)
    amplitude spectrum: C₀ = 2, C₂ = 5, C₃ = 2, C₇ = 1 (all others 0)
(b) ω₀ = 1.  c₀ = 2;  c±₁ = ¼e^{±j45°};  c±₃ = 1;  −sin(4t + 30°) = cos(4t + 120°) ⇒ c±₄ =
¼e^{±j120°}
    |c|: 2 (k=0), ¼ (±1), 1 (±3), ½ (±4);   ∠c: ±45° (k = ±1), 0 (±3), ±120° (±4)
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(c) cos $2\pi f_0 t = \frac{1}{2}e^{j2\pi f_0 t} + \frac{1}{2}e^{-j2\pi f_0 t} \Rightarrow a_{\pm 1} = \frac{1}{2}$ (**real**). sin $2\pi f_0 t = (1/2j)e^{j2\pi f_0 t} - (1/2j)e^{-j2\pi f_0 t} \Rightarrow a_{\pm 1} = \mp j/2$ (**purely imaginary**). Both satisfy Dirichlet (bounded, smooth, one max/min per period). In general a_k needs a full period; for even signals $a_k = (2/T)\int_0^{T/2} x \cos k\omega_0 t\, dt$ and for odd signals $a_k = (-2j/T)\int_0^{T/2} x \sin k\omega_0 t\, dt$ — half a cycle suffices (Lesson 13).

(d) x(t) = A(2|t| − 1), |t| ≤ 1, period 2 ($\omega_0 = \pi$). Even ⇒ $b_n = 0$. a_0 = average = 0 (equal areas above and below).

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aₙ = (2/2)∫_{−1}^{1} A(2|t| − 1)cos nπt dt = 2A∫₀¹(2t − 1)cos nπt dt = 4A[(−1)ⁿ − 1]/(n²π²)
   = −8A/(n²π²) for odd n,  0 for even n
x(t) = −(8A/π²)[cos πt + (1/9)cos 3πt + (1/25)cos 5πt + …]
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(e)(i)  ω₀ = 2: a₀ = 3;  a₂ = cos π/4 = 0.707, b₂ = −sin π/4 = −0.707;
        sin(10t + π/3) = (√3/2)cos10t + ½ sin10t ⇒ a₅ = 0.866, b₅ = 0.5
(ii)    ω₀ = 2, even about π/2: c₀ = 1;  c_k = −4/(π²k²) for odd k, 0 for even k ≠ 0
        x = 1 − (8/π²)Σ_{odd} cos(2kt)/k²
(iii)   Cₙ = (Kτ/T)·sin(nπτ/T)/(nπτ/T); τ = 1: T = 5, 10, 15 ⇒ line spacing 2π/T shrinks and
every line
        shrinks (∝ 1/T) while the sinc envelope (zeros at multiples of 2π/τ) stays fixed
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Lesson 13 — Linearity, Shift, Reversal, Scaling and Symmetry

Slides 2–35 (Lec 6) · symmetry rules: textbook

Property	Signal (same T)	Coefficients
Linearity (sl. 5–8)	$\alpha x + \beta y$	$\alpha a_k + \beta b_k$
Time shift (sl. 9–17)	$x(t - t_0)$	$a_k e^{-jk\omega_0 t_0}$ (magnitude unchanged)
Reversal (sl. 18–23)	$x(-t)$	a_{-k}
Scaling (sl. 27–33)	$x(\alpha t)$, $\alpha > 0$	a_k, but $\omega_0 \to \alpha\omega_0$ (period T/α)
Conjugation	$x^*(t)$	a^*_{-k}

Worked (sl. 24–26, 34–35): $x(4 - t) + y(t - 3) \leftrightarrow a_{-k}e^{-j4k\omega_0} + b_k e^{-j3k\omega_0}$; $x(4 - 3t) \leftrightarrow a_{-k}e^{-j4k\omega_0}$ with $\omega_0' = 3\omega_0$ (shift, then reverse, then scale — the coefficient does not change in the last step).

TRAP Slide 26 writes $y(t - 3) \leftrightarrow a_k e^{-j3k\omega_0}$; it must be **$b_k$** $e^{-j3k\omega_0}$ (y has coefficients b_k), as the boxed final answer shows.

Symmetry Rules (textbook)

Signal	Coefficients
real	$a_{-k} = a_k^*$ ($\lvert a_k\rvert$ even, $\angle a_k$ odd)
real and even	a_k real and even (cosine terms only)
real and odd	a_k purely imaginary and odd, $a_0 = 0$ (sine terms only)
half-wave symmetric: $x(t + T/2) = -x(t)$	$a_k = 0$ for every even k (odd harmonics only)

Proof of even → real: $a_k = (1/T)\int_{-T/2}^{T/2} x(t)[\cos k\omega_0 t - j \sin k\omega_0 t]dt$; $x \cdot \sin$ is odd, so its integral vanishes and $a_k = (2/T)\int_0^{T/2} x \cos k\omega_0 t\, dt$ is real. For odd x the cosine part vanishes: $a_k = -(2j/T)\int_0^{T/2} x \sin k\omega_0 t\, dt$, imaginary. Both need only half a cycle.

Exam Approach

1. Write the new signal as a chain of operations on the known one.
2. Apply the rules in the same order (shift → reverse → scale), updating ω_0 after scaling.
3. Use symmetry to predict which coefficients vanish and whether they are real/imaginary — a free check.

Practice — Lesson 13

(a) Actual — CSE 219 2023-24 Q4. (b) Actual — CSE 219 2022-23 Q5(d). (c) Actual — CSE 311 2022-23 Q1(a), 2018-19 Q2(b). (d) Actual — EEE 301 2011-12 Q1(a).

(a) $x(t) = A$ $(0 < t < T_0/2)$, $-A$ $(-T_0/2 < t < 0)$, periodic; $y(t) = x(t - T_0/6)$. (a) Show y has half-wave symmetry. (b) Find Y_k from X_k. (c) Magnitude/phase spectra up to $k = 5$. (d) Compare the trigonometric series of x and y. (5 + 10 + 10 + 5)

(b) Is $x(t) = \Sigma_{k=0}^{100}(½)^k e^{jk2\pi t/50}$ real-valued? even? (5)

(c) Show that the exponential FS coefficients of an even (odd) periodic signal are real (imaginary) and can be computed over half a cycle. (15–17)

(d) x is real and odd, $T = 2$, $c_k = 0$ for $|k| > 1$, $½\int_0^2 |x|^2 dt = 1$. Give two signals that satisfy this. (18)

✓ Solution

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(a) X_k = (1/T₀)[∫₀^{T₀/2} A e^{-jkω₀t}dt − ∫_{-T₀/2}^{0} A e^{-jkω₀t}dt] = (A/(jkπ))[1 − (−1)^k]
    X_k = 2A/(jπk) for odd k, 0 for even k (half-wave symmetric, real-odd ⇒ imaginary)
    (a) y(t + T₀/2) = x(t + T₀/2 − T₀/6) = −x(t − T₀/6) = −y(t)  ✓ (a shift keeps half-wave symmetry)
    (b) Y_k = X_k e^{-jkω₀T₀/6} = X_k e^{-jkπ/3} = (2A/(jπk)) e^{-jkπ/3}, odd k; 0 for even k
    (c) |Y_k| = 2A/(π|k|): k = ±1: 0.637A, ±3: 0.212A, ±5: 0.127A
        ∠Y_k = −π/2 − kπ/3 (k > 0): k=1: −5π/6;  k=3: −3π/2 ≡ +π/2;  k=5: −13π/6 ≡ −π/6  (odd in k)
    (d) x(t) = Σ_{odd k} (4A/(πk)) sin kω₀t;   y(t) = Σ_{odd k} (4A/(πk)) sin(kω₀t − kπ/3)
        same amplitudes (only odd harmonics); each harmonic phase-shifted by −kπ/3 (∝ k: pure delay)
```

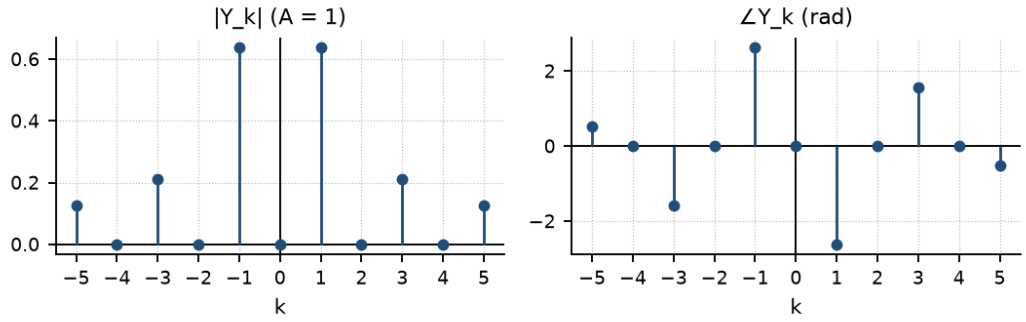


Figure 13.1 — Spectra of y(t) = x(t − T_0/6) for CSE 219 2023-24 Q4 (A = 1)

(b) Real requires $a_{-k} = a_k$. *Here a_k = (½)^k for 0 ≤ k ≤ 100 but a_{-k} = 0 ≠ a_k* ⇒ **not real**. Even requires $a_{-k} = a_k$ ⇒ **not even** either (x is a sum of positive-frequency exponentials only).

(c) Proof in the box above (split $e^{-jk\omega_0 t}$ into cos − j sin and use odd × even = odd).

(d) Real odd with only k = ±1 ⇒ x = B sin πt ($\omega_0 = \pi$). Power $\frac{1}{2}\int_0^2 B^2\sin^2\pi t\,dt = B^2/2 = 1 \Rightarrow B = \pm\sqrt{2}$: $x_1 = \sqrt{2}\sin\pi t$ and $x_2 = -\sqrt{2}\sin\pi t$.

Lesson 14 — Differentiation, Impulses, Impulse Trains, Integration

Slides 36–72 (Lec 6)

$dx/dt \leftrightarrow jk\omega_0 a_k$ $\int x\,dt \leftrightarrow a_k/(jk\omega_0)$ ($k \neq 0$; find a_0 separately)

Every harmonic is a rotating arrow; differentiating gives its velocity: speed $|k\omega_0||a_k|$, direction tangent — multiplication by j (sl. 37–39). Example: $d/dt\ x(4-3t) \leftrightarrow j3k\omega_0\, a_{-k}\, e^{-j4k\omega_0}$ (use the current $\omega_0' = 3\omega_0$).

Derivative sketch (sl. 42–44): slopes between jumps + impulses equal to each jump. **Sifting** (sl. 45–51): $\int e^t\delta(t)dt = 1$; $\int_1^{\infty} e^t\delta(t)dt = 0$ (outside); $\int_{-5}^{5} e^t[\delta(t+1) + 2\delta(t-2)]dt = e^{-1} + 2e^2$.

Impulse Train (sl. 52–56)

$\Sigma_m\, \delta(t - mT) \leftrightarrow a_k = 1/T$ for every k **trig form: $1/T + (2/T)\Sigma \cos(2\pi nt/T)$**

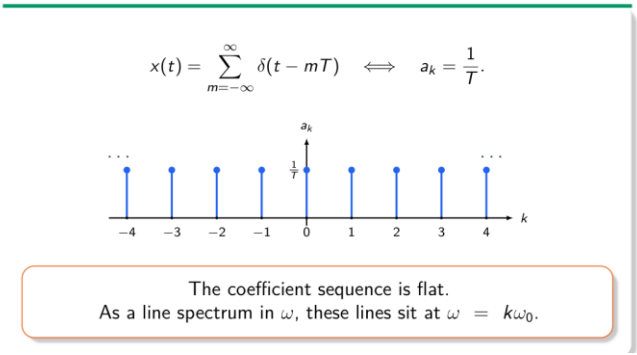


Figure 14.1 — An impulse train has a flat line spectrum (Lec 6, sl. 55)

Two Worked Uses (sl. 57–72)

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Rectangular pulse train: dx/dt = δ(t + T₁) − δ(t − T₁) per period ⇒ b_k = (1/T)(e^{jkω₀T₁} − e^{−jkω₀T₁})
   = (2j/T) sin kω₀T₁ = jkω₀ · sin(kω₀T₁)/(kπ) ✓
Sawtooth x = t (0 ≤ t < 1), T = 1: dx/dt = 1 − Σδ(t − m) ⇒ b₀ = 0, b_k = −1
   a_k = b_k/(jkω₀) = −1/(j2πk) = j/(2πk),  a₀ = ∫₀¹ t dt = ½
```

Exam Approach

1. Piecewise-constant or piecewise-linear x: differentiate until only impulses remain.
2. Impulse coefficients by sifting: $\delta(t - t_0)$ in one period $\leftrightarrow (1/T)e^{-jk\omega_0 t_0}$.
3. Divide by $(jk\omega_0)^m$; get a_0 from the average.
4. Trains with alternating signs = sum of shifted trains.

Practice — Lesson 14

(a) Actual — EEE 211 2022-23 Q8(a). (b) Actual — EEE 211 2019-20 Q7(a). (c) Actual — EEE 211 2020-21 Q7(a). (d) Actual — EEE 301 2014-15 Q4(a), 2015-16 Q6(a).

(a) Trigonometric FS ($A_0 + \Sigma A_n \cos(2\pi nt/T_0 + \varphi)$) of $\delta_{T_0}(t)$; amplitude spectrum. (20)

(b) Exponential FS of (i) unit impulses at every integer; (ii) impulses −1 at even integers and +1 at odd integers. (15)

(c) x(t) = 1 − |t| on [−2, 2], period 4 (triangle between ±1); y(t) = 4 on (0, 2), 0 on (2, 4), period 4. Find the exponential FS of x, then y by properties; amplitude spectra. (20)

(d) C_n = ½[1 − (−1)^n], T = 20 ms: real signal? find x(t). Impulse train: FS, FT, PSD.

✓ Solution

(a) a_k = 1/T_0 for all k ⇒ $\delta_{T_0}(t)$ = 1/T_0 + (2/T_0)$\Sigma_{n\geq 1}$ cos(2πnt/T_0): A_0 = 1/T_0, A_n = 2/T_0, φ = 0. Amplitude spectrum: a line of height 1/T_0 at n = 0 and 2/T_0 at every harmonic n·(1/T_0) Hz.

(b) (i) T = 1: a_k = 1 for all k ⇒ x = Σ $e^{j2\pi kt}$. (ii) T = 2, one period has −δ(t) and +δ(t − 1): a_k = ½[−1 + $e^{-jk\pi}$] = ½[(−1)^k − 1] = −1 for odd k, 0 for even k ⇒ x = $-\Sigma_{\text{odd k}}$ $e^{jk\pi t}$ = $-2\Sigma_{\text{odd}}$cos kπt.

```
(c) x′ = −1 on (0,2), +1 on (−2,0); x″ = 2δ(t + 2) − 2δ(t) per period (T = 4, ω₀ = π/2)
    (jkω₀)² a_k = (1/4)[2e^{jkπ} − 2] ⇒ a_k = (1 − (−1)^k)/(2·k²π²/4) = 4/(k²π²) for odd k, 0
even; a₀ = 0
    y = 2 − 2x′(t)  (x′ = −1 on (0,2) gives 4, +1 on (2,4) gives 0)
    b₀ = 2;  b_k = −2·jkω₀ a_k = −2j(kπ/2)(4/(k²π²)) = −4j/(πk) for odd k, 0 for even k ≠ 0
    |a_k| = 4/(π²k²): 0.405, 0.045, 0.016 (k = 1, 3, 5);   |b_k| = 4/(π|k|): 1.273, 0.424,
0.255; |b₀| = 2
```

(d) C_n = 1 for odd n, 0 for even n (C_0 = 0). C_{-n} = C_n* and real ⇒ x is real and even. x = $\Sigma_{\text{odd n}}$ $e^{jn\omega_0 t}$, ω_0 = 2π/0.02 = 100π rad/s. Since $\Sigma_{\text{all}} e^{jn\omega_0 t}$ = $T_0\Sigma\delta(t - kT_0)$ and the even-n part is (T_0/2)Σδ(t − kT_0/2), x(t) = (T_0/2)Σ_k(−1)^kδ(t − kT_0/2) = 0.01Σ(−1)^kδ(t − 0.01k): alternating impulses every 10 ms. Impulse train δ_T: FS a_k = 1/T; FT (2π/T)Σδ(ω − 2πk/T); PSD S(ω) = (2π/T²)Σδ(ω − 2πk/T).

Lesson 15 — Multiplication, Parseval, and Periodic Inputs to LTI Systems

Slides 73–81 (Lec 6) · Parseval and LTI response: textbook

$$x(t)y(t) \leftrightarrow c_k = \sum_p a_p b_{k-p} \quad \text{(convolution of coefficients)}$$

Proof (sl. 75–78): put $z = e^{j\omega_0 t}$; each series is a polynomial in z, and the product's z^k coefficient collects pairs with p + q = k. Example: $x = 1 + \cos \omega_0 t$ ($a_0 = 1$, $a_{\pm 1} = ½$) ⇒ $x^2 \leftrightarrow \{¼, 1, 3/2, 1, ¼\}$ for k = −2…2; check: $(1 + \cos)^2 = 3/2 + 2\cos \omega_0 t + ½\cos 2\omega_0 t$ ✓.

Parseval and Power (textbook)

$$P = (1/T)\int_T |x|^2 dt = \sum |a_k|^2 \quad (= C_0^2 + \sum C_n^2/2 \text{ in compact trig form})$$

Periodic Input to an LTI System (textbook)

$e^{jk\omega_0 t}$ is an eigenfunction: it comes out as $H(jk\omega_0)e^{jk\omega_0 t}$. So a periodic input with coefficients a_k gives a periodic output (same T) with

$$y_k = H(jk\omega_0)\, a_k \qquad A\cos(\omega t + \theta) \rightarrow A|H(j\omega)|\cos(\omega t + \theta + \angle H(j\omega))$$

MEMORIZE Distortionless system (EEE 2017-18 Q1(b)): $H = K e^{-j\omega t_d} \Rightarrow y_k = K e^{-jk\omega_0 t_d} a_k$ (same magnitudes ×K, phase linear in k). Ideal LPF with cutoff ω_c: keep only harmonics with $|k\omega_0| < \omega_c$.

Exam Approach

1. Find a_k of the input (inspection / standard waveform).
2. Find H(jω) of the system (circuit: voltage divider with jωL, 1/(jωC); ODE: replace d/dt by jω).
3. Multiply harmonic by harmonic; list magnitudes and phases.
4. Power: Parseval on the output coefficients.

Practice — Lesson 15

(a) Actual — EEE 211 2023-24 Q7(b). (b) Actual — EEE 211 2020-21 Q7(b). (c) Actual — EEE 211 2019-20 Q5(b). (d) Actual — EEE 211 2021-22 Q3(a). (e) Older: EEE 211 2017-18 Q1(c), EEE 301 2015-16 Q4(b), 2011-12 Q1(b), 2012-13 Q4(b), EEE 211 2018-19 Q3(c).

(a) Half-wave rectified sine x(t) = sin(2πt/T) on the first half period, T = 0.5 ms, feeds (1) a differentiator, (2) a LPF with $|H_2| = 5$, linear phase (±π/2 at ±f_c), f_c = 1 kHz. (i) FS coefficients of y_1 from those of x. (ii) $y_2(t)$; which harmonics survive? (iii) Compare the two systems. (25)

(b) y″ + 3y′ + 2y = x′ + x, x = 2cos t + 6cos 3t: amplitude and phase spectra of y; average power by Parseval. (15)

(c) Full-wave rectified sin t feeds an RC low-pass (R = 15 Ω, C = 1/5 F, output across C). H(ω) and y(t). (20)

(d) Period 4: x = δ(t − 1) + 2[u(t − 2) − u(t − 4)] in one period, applied to a 0.1 H series inductor, 100 μF shunt capacitor and 1 kΩ load. Magnitude and phase spectra of x and y. (20)

(e) Older (see solutions).

✓ Solution

(a) $f_0 = 2$ kHz, $\omega_0 = 4000\pi$. Half-wave rectified sine (amplitude 1):
$a_0 = 1/\pi$, $a_{\pm 1} = \mp j/4$, $a_k = -1/(\pi(k^2 - 1))$ for even k, 0 for odd $|k| \geq 3$
(i) Differentiator $H_1 = j\omega$: $b_k = jk\omega_0 a_k \Rightarrow b_0 = 0$, $b_{\pm 1} = (j\omega_0)(\mp j/4)\cdot(\pm 1) = \omega_0/4 = 1000\pi$,
$b_k = -jk\omega_0/(\pi(k^2 - 1)) = -j4000k/(k^2 - 1)$ for even k
(ii) LPF passes $|f| < 1$ kHz only: the lowest harmonic is 2 kHz ⇒ only DC survives:
$y_2(t) = H_2(0)a_0 = 5/\pi \approx 1.59$ (constant; harmonic $k = 0$ only)
(iii) the differentiator scales harmonic k by $|k|\omega_0$ – it boosts high frequencies and removes DC;
the LPF removes everything except DC – it keeps the average and rejects all harmonics

(b) $H(j\omega) = (j\omega + 1)/((j\omega)^2 + 3j\omega + 2) = 1/(j\omega + 2)$
$\omega = 1$: $|H| = 1/\sqrt{5}$, $\angle H = -26.57°$; $\omega = 3$: $|H| = 1/\sqrt{13}$, $\angle H = -56.31°$
$y = (2/\sqrt{5})\cos(t - 26.57°) + (6/\sqrt{13})\cos(3t - 56.31°) = 0.894\cos(t - 26.57°) + 1.664\cos(3t - 56.31°)$
$P = \frac{1}{2}(0.894^2 + 1.664^2) = \frac{1}{2}(4/5 + 36/13) = 1.785$ W (1-Ω basis)
(c) $|\sin t| = 2/\pi - (4/\pi)\Sigma_{k\geq 1} \cos 2kt/(4k^2 - 1)$ ($T = \pi$, $\omega_0 = 2$)
$RC = 3$: $H(\omega) = 1/(1 + j3\omega)$
$y(t) = 2/\pi - (4/\pi)\Sigma\, |H(j2k)| \cos(2kt - \tan^{-1}6k)/(4k^2 - 1)$
$= 0.637 - 0.0698\cos(2t - 80.54°) - 0.0070\cos(4t - 85.24°) - \ldots$ (nearly pure DC)
(d) $a_0 = (1 + 4)/4 = 5/4$; $a_k = \frac{1}{4}e^{-jk\pi/2} + j(1 - (-1)^k)/(k\pi)$ ($\omega_0 = \pi/2$)
$a_{\pm 1} = \pm j(1/\pi\cdot 2 - \frac{1}{4}) = \pm j0.387$; $a_{\pm 2} = -\frac{1}{4}$; $a_{\pm 3} = \pm j(2/(3\pi) + \frac{1}{4}) = \pm j0.462$; $a_4 = \frac{1}{4}$ …
$H(j\omega) = 1/(1 - \omega^2 LC + j\omega L/R)$, $LC = 10^{-5}$, $L/R = 10^{-4} \Rightarrow |H(jk\pi/2)| = 1.000$, $\angle H \approx -0.01°$ for low k
⇒ output spectrum ≈ input spectrum (the filter's corner, $1/\sqrt{LC} = 316$ rad/s, is far above $k\omega_0$ for small k)

(e) EEE 211 2017-18 Q1(c): H passes $|\omega| \leq 3\pi \Rightarrow y = 1 + 2\cos 2\pi t$ ($\cos 4\pi t$, $\cos 6\pi t$ removed)
EEE 301 2015-16 Q4(b): $a_{\pm 1} = \frac{1}{2}$, $b_{\pm 1} = \pm 1/(2j) \Rightarrow c_0 = 0$, $c_2 = 1/(4j)$, $c_{-2} = -1/(4j) \Rightarrow z = \frac{1}{2}\sin 8\pi t$
EEE 301 2011-12 Q1(b): x = 2 + square wave (±1, T = 8, $\omega_0 = \pi/4$); H(0) = 4; odd harmonics at $k\pi/4$ meet
$H = \sin(k\pi)/(k\pi/4) = 0 \Rightarrow y(t) = 4\cdot 2 = 8$ (constant)
EEE 301 2012-13 Q4(b): half-wave cos t: 2nd and 4th harmonic amplitudes $2/(3\pi) = 0.212$, $2/(15\pi) = 0.042$;
the LC–R network has $|H| \approx 1$ at $\omega = 2, 4$ ($LC = 10^{-5}$) ⇒ output amplitudes ≈ 0.212 and 0.042
EEE 211 2018-19 Q3(c): $T = 2$, $\omega_0 = \pi$, $C_n = 1$ (odd n); the band $2\pi < |\omega| < 4\pi$ passes only $n = \pm 3 \Rightarrow$
$y = 2\cos(3\pi t - \pi/2) = 2\sin 3\pi t$ (with $-\pi/2$ for $\omega > 0$); h(t) = band-pass with $-j$ sgn ω (method)

Quick Notes — Fourier Series

Revision points for Lessons 11–15.

L11 · Idea, Orthogonality

- $x = \Sigma\, a_k\, e^{jk\omega_0 t}$, $a_k = (1/T)\int_T x\, e^{-jk\omega_0 t} dt$, a_0 = average.
- Orthogonal: $\int \varphi_i \varphi_j^* = 0$; normalize by $1/\sqrt{E}$; $c_i = \langle x, \varphi_i \rangle / E_i$.
- Dirichlet: absolutely integrable, finite extrema, finite jumps — sufficient, not necessary.

L12 · Coefficients

- Square wave: $a_0 = 2T_1/T$, $a_k = \sin(k\omega_0 T_1)/(k\pi)$.
- Ramp ($T = 3$): $a_0 = ½$, $a_k = (e^{-jk\omega_0} - 1)/(2k^2\omega_0^2)$ (slide prints 6).
- $c_n = (a_n - jb_n)/2$, $C_n = 2|c_n|$.

L13 · Properties

- Shift: $a_k\, e^{-jk\omega_0 t_0}$; reverse: a_{-k}; $x(\alpha t)$: same a_k, $\omega_0 \to \alpha\omega_0$.
- Real: $a_{-k} = a_k^*$; even → real; odd → imaginary; half-wave → odd harmonics only.

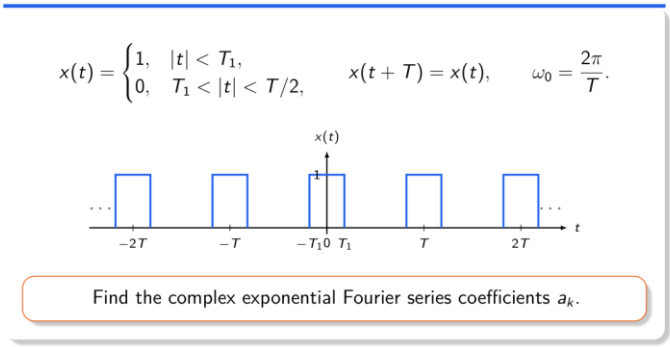
L14 · Derivative, Trains

- $dx/dt \leftrightarrow jk\omega_0 a_k$; $\int \leftrightarrow a_k/(jk\omega_0)$ (a_0 separately).
- $\Sigma\delta(t - mT) \leftrightarrow 1/T$. Sawtooth t ($T = 1$): $a_0 = ½$, $a_k = j/(2\pi k)$.

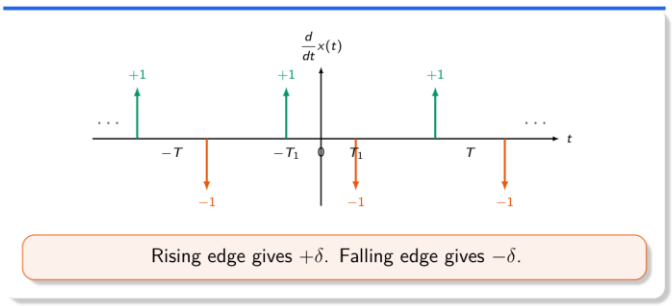
L15 · Products, LTI

- $xy \leftrightarrow \Sigma\, a_p b_{k-p}$. $P = \Sigma|a_k|^2$.
- LTI: $y_k = H(jk\omega_0) a_k$.

Key Diagrams to Reproduce



The periodic square wave (Lec 5, sl. 30)



Derivative of the rectangular pulse train (Lec 6, sl. 59)